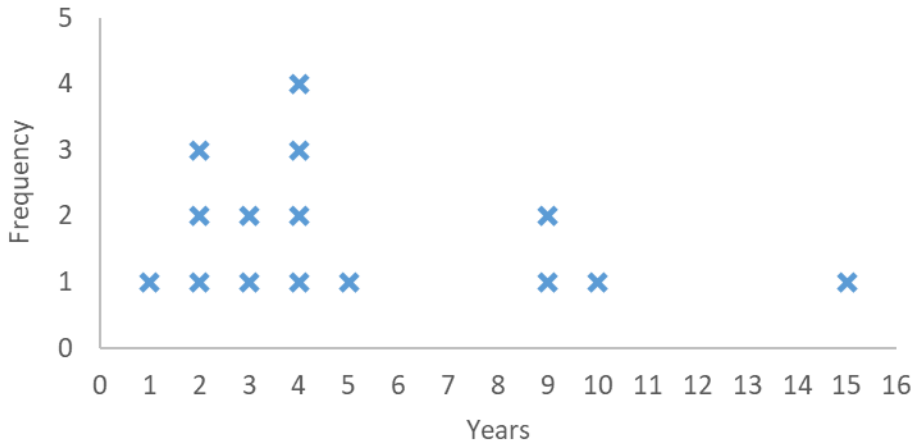
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EXERCISE SHEET / THEME 4: RANDOM VARIABLES AND DISTRIBUTION FUNCTIONS.

EXERCISE 1. WORK EXPERIENCE DOT PLOT.

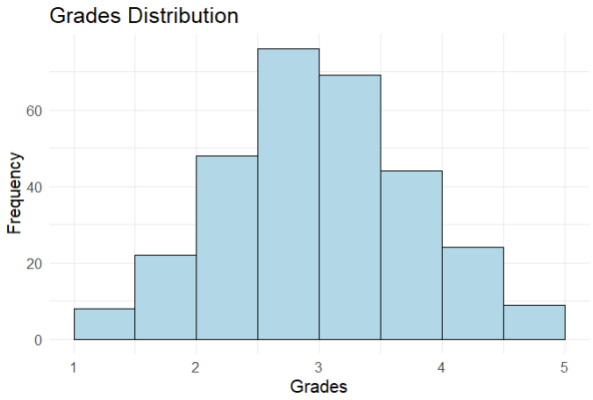
Context: You are analyzing the background of a group of master students. You receive the following dot plot that displays the distribution of their years of work experience.

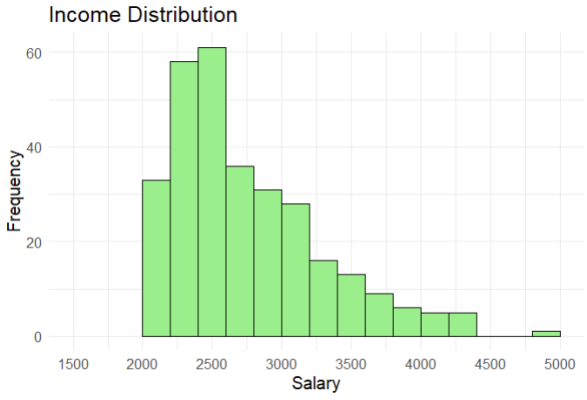


a) What could you say about it? You can refer to concepts as symmetry/asymmetry, clusters, gaps, outliers within the distribution.
b) Calculate the median and the mean.

EXERCISE 2. GRADES AND SALARY HISTOGRAMS.

Context: After two years, the students finish their program. Now you receive the following histograms displaying the distribution of both: their final grades and their salaries in their first job after their studies.





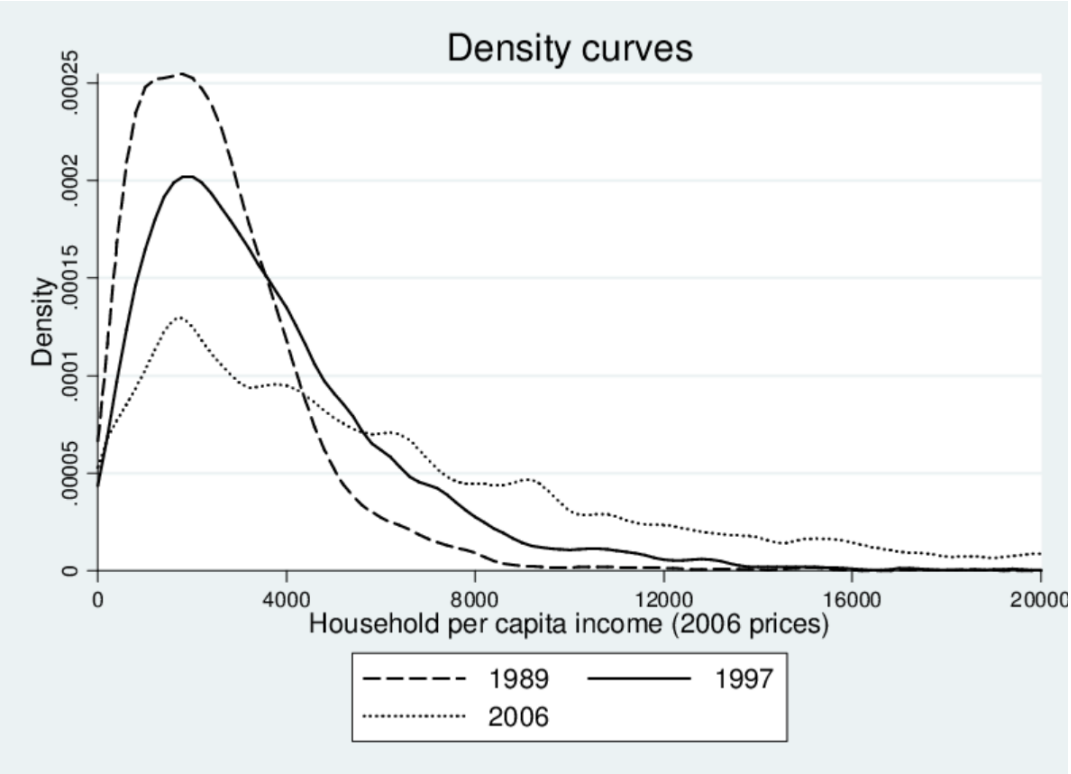
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Compare the two distributions.

a) Refer to the symmetry/asymmetry.
b) Refer to the skewness.
c) What is the mode of each distribution (which range)?
d) Without calculating them, are the median and the mean close to each other in both cases?
e) Are there any outliers? Can they affect the median or the mean?

EXERCISE 3. HOUSEHOLDS INCOME DENSITY CURVES.

Context: With the first two exercises you get a feeling about the distributions displayed by a dot plot and a histogram. For this exercise you will analyze another graph used for this: the density curve. Even though it looks slightly different, it also shows distribution (the curve will be higher where there are more observations and lower where there are fewer). This plot shows the household income per capita in a region in three different periods (1989, 1997 and 2006).



Source: Bonnefond, C., & Clément, M. (2012). An analysis of income polarisation in rural and urban China. Post-Communist Economies, 24(1), 15–37. https://doi.org/10.1080/14631377.2012.647972

Answer the following questions

a) In which period was more share of the population with an income per capita of 4000 or less?

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b) Which period shows more equal incomes? Explain why.